LAB ASSIGNMENT – SQL FUNCTIONS

APPLY SQL FUNCTIONS: STRING, NUMERIC, DATE, AND AGGREGATE FUNCTIONS

Name: Harshitha J
USN: 1RUA25BCA0041
Course: BCA

1. AIM

To apply and demonstrate SQL String, Numeric, Date, and Aggregate Functions using a School Student Database.

2. OBJECTIVE

The objectives of this assignment are:

1. To understand the concept of SQL Functions.
2. To apply String Functions for manipulating text data.
3. To apply Numeric Functions for performing calculations.
4. To apply Date Functions for working with date values.
5. To apply Aggregate Functions for performing calculations on multiple rows.
6. To use the GROUP BY clause with aggregate functions.
7. To combine multiple SQL functions in a single query.

SQL functions accept input, perform an operation, and return an output.

3. DATABASE SCENARIO

A school database is used to maintain basic information about its students.

The database contains a table named **STUDENT** with information such as Student ID, Student Name, Class, Marks, Fee Paid, Date of Birth, and Admission Date.

4. CREATION OF STUDENT TABLE

INSERTING RECORDS INTO STUDENT TABLE

DISPLAYING THE STUDENT TABLE



PART A – STRING FUNCTIONS

String functions are used to manipulate text data. In this table, Student_Name and Class are examples of text data

UPPER() FUNCTION

Purpose

The UPPER() function converts text into uppercase letters



LOWER() FUNCTION

Purpose

The LOWER() function converts text into lowercase letters.



LENGTH() FUNCTION

Purpose

The LENGTH() function finds the number of characters in a string.



CONCAT() FUNCTION

Purpose

The CONCAT() function joins two or more strings together.



SUBSTRING() FUNCTION

Purpose

The SUBSTRING() function extracts a specific part of a string.



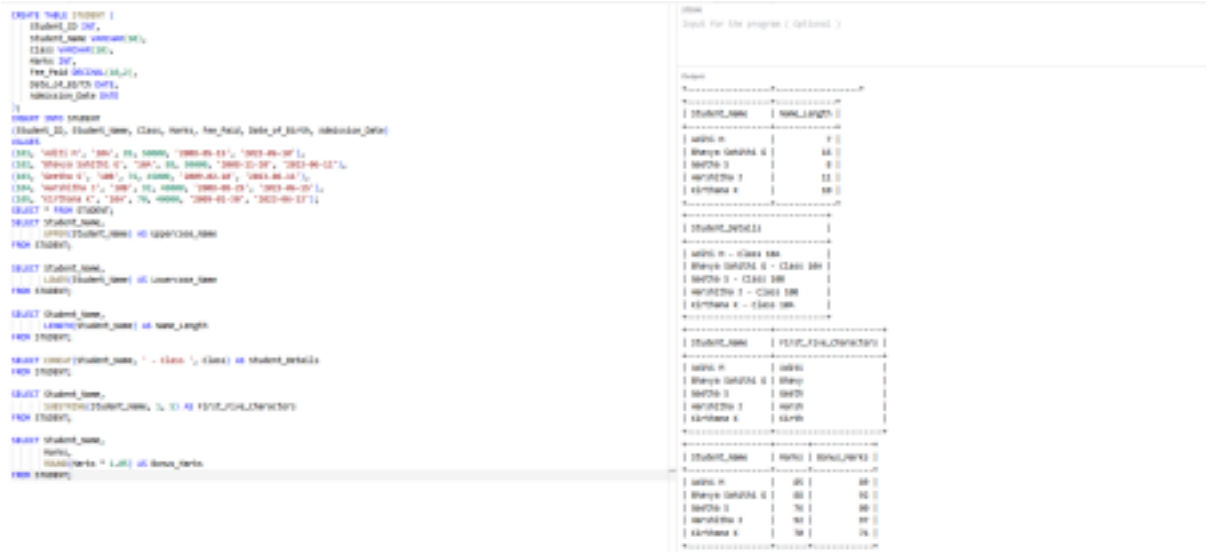
PART B – NUMERIC FUNCTIONS

Numeric functions perform calculations on numerical data such as Marks and

Fee_Paid. ROUND() FUNCTION

Purpose

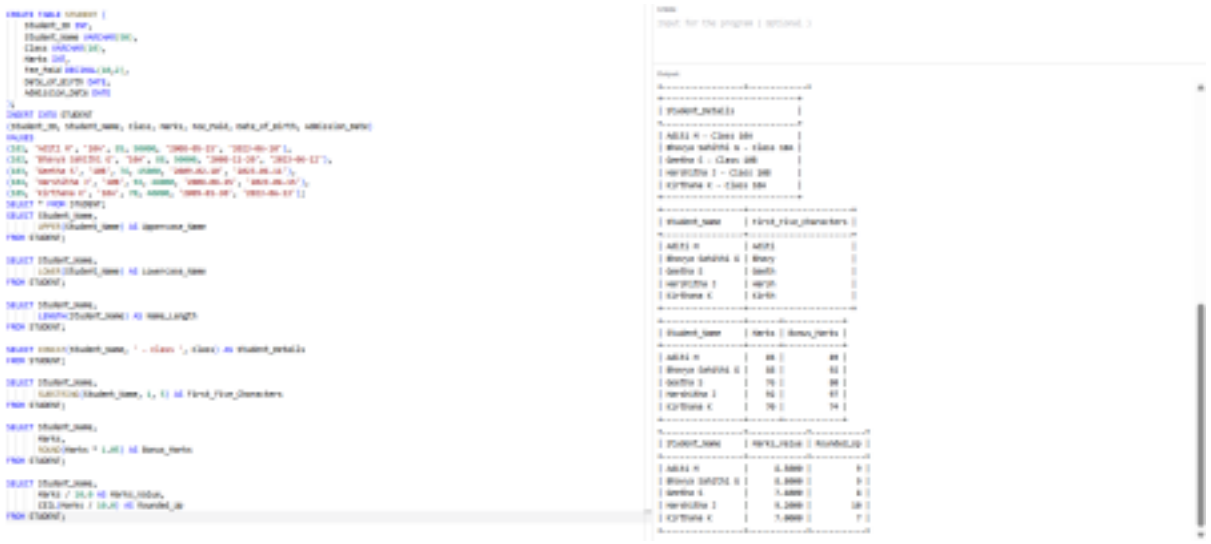
The ROUND() function rounds a numerical value to the nearest integer.



CEIL() FUNCTION

Purpose

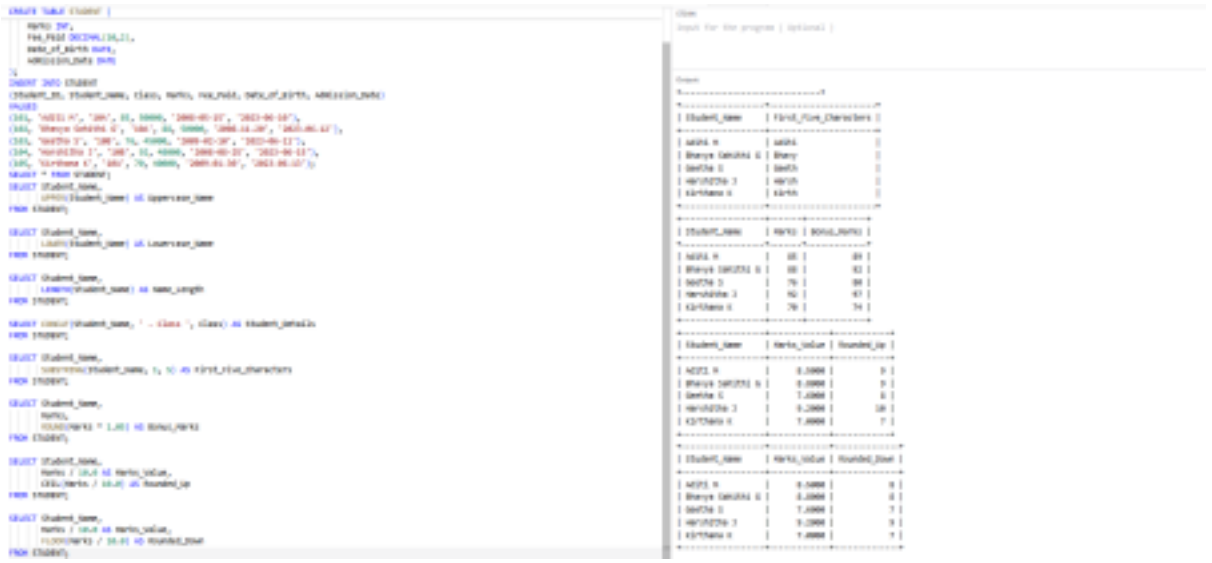
The CEIL() function rounds a number upward to the next integer.



FLOOR() FUNCTION

Purpose

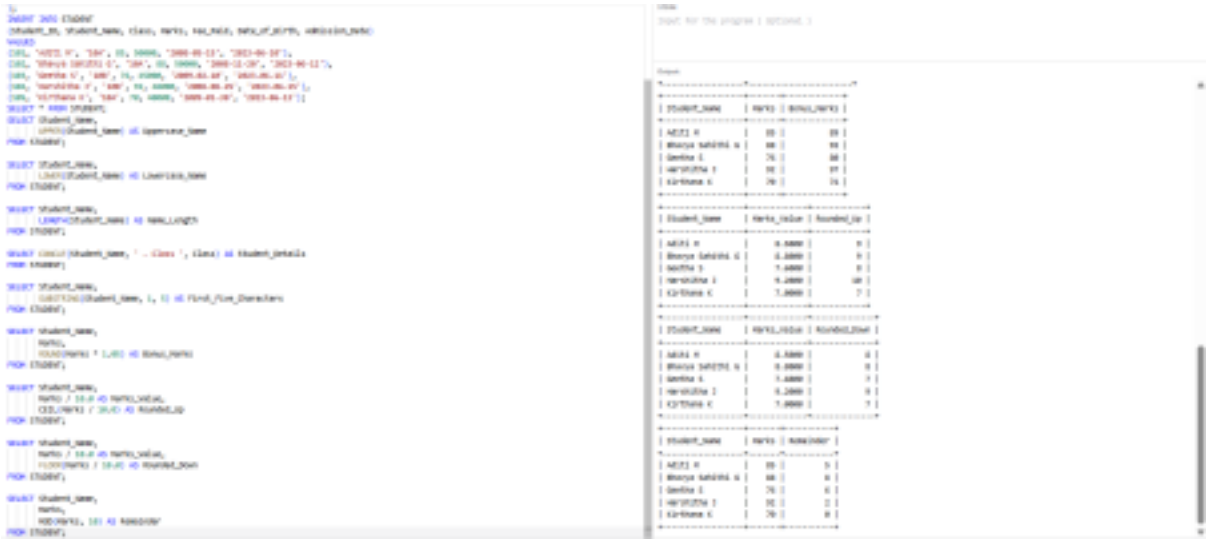
The FLOOR() function rounds a number downward to the nearest integer.



MOD() FUNCTION

Purpose

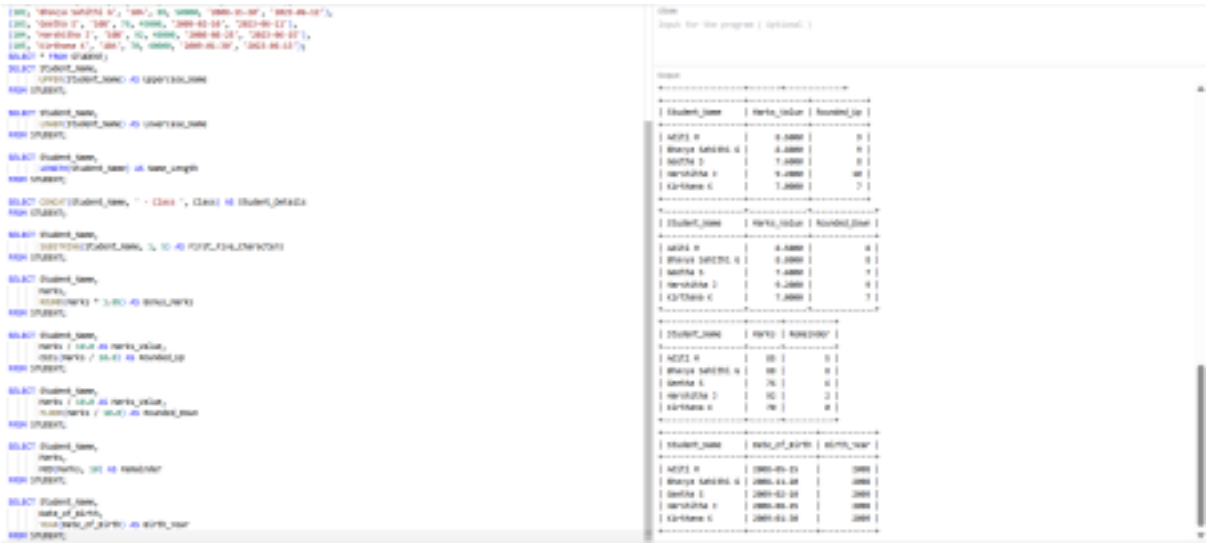
The MOD() function returns the remainder after division.



ABS() FUNCTION

Purpose

The ABS() function returns the absolute value of a number. It can be used to find the difference between marks and a target value.



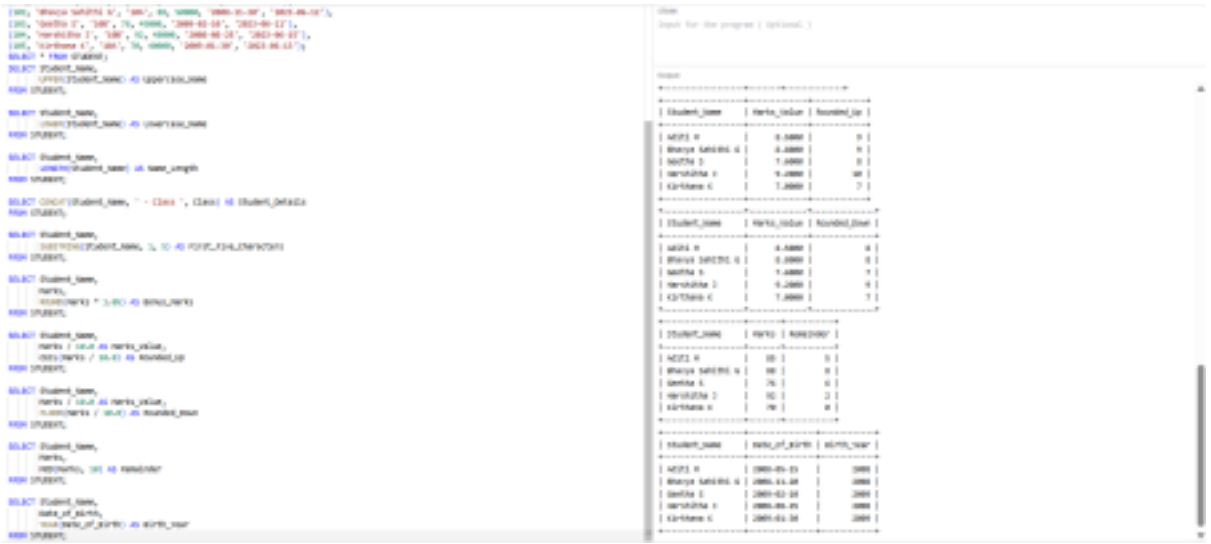
PART C – DATE FUNCTIONS

Date functions are used to work with date values such as Date of Birth and Admission Date.

YEAR() FUNCTION

Purpose

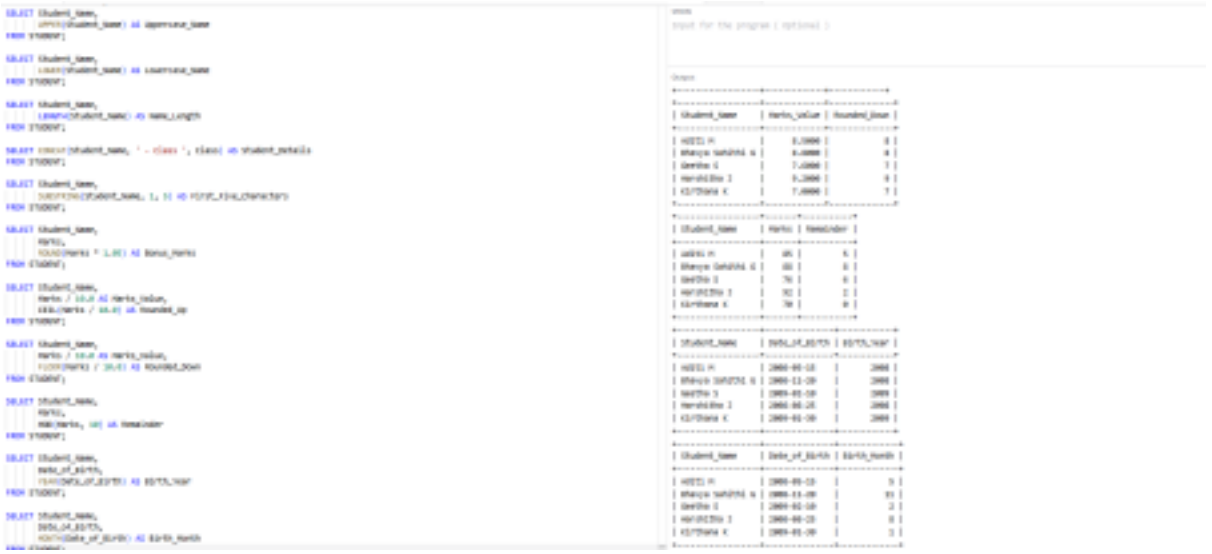
The YEAR() function extracts the year from a date.



MONTH() FUNCTION

Purpose

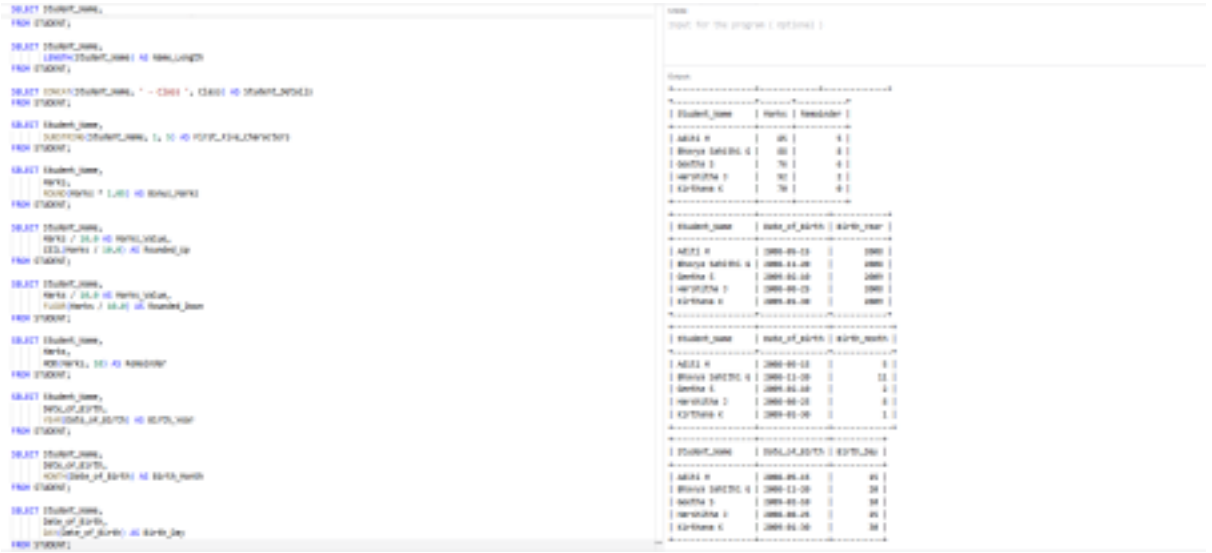
The MONTH() function extracts the month from a date.



DAY() FUNCTION

Purpose

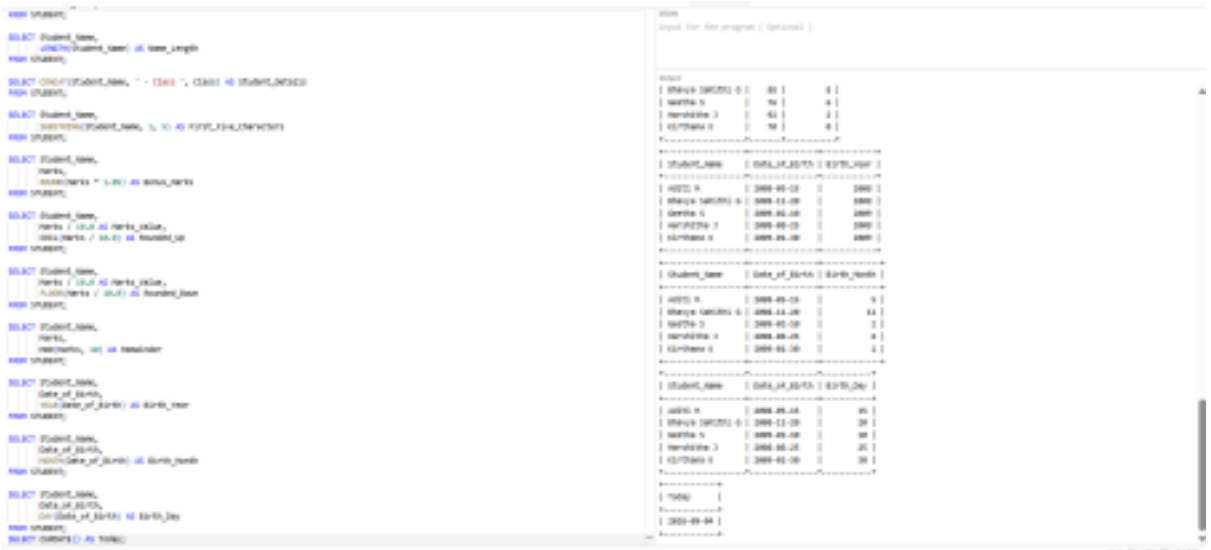
The DAY() function extracts the day from a date.



TIMESTAMPDIFF() FUNCTION

Purpose

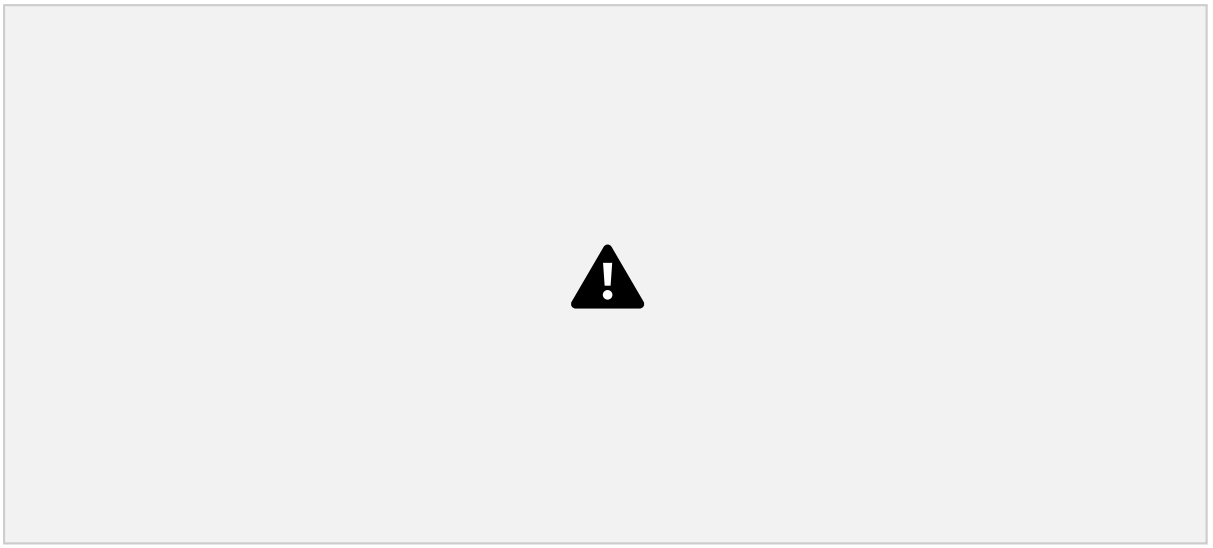
The TIMESTAMPDIFF() function calculates the difference between two dates. It can be used to calculate the age of students.



CURDATE() FUNCTION

Purpose

The CURDATE() function returns the current date.



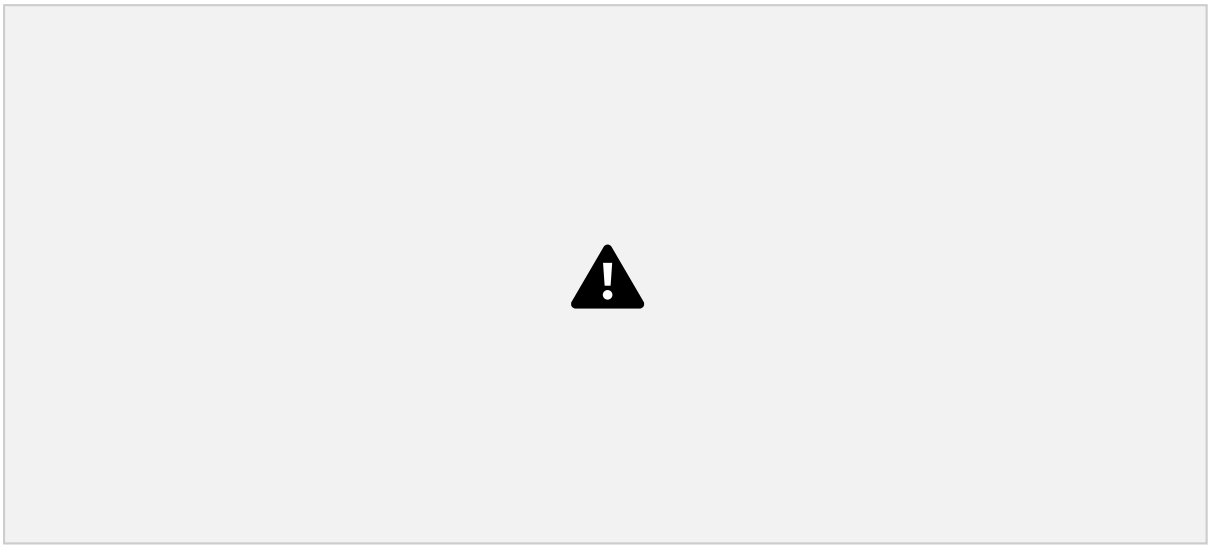
PART D – AGGREGATE FUNCTIONS

Aggregate functions operate on multiple rows together and produce a single result.

COUNT() FUNCTION

Purpose

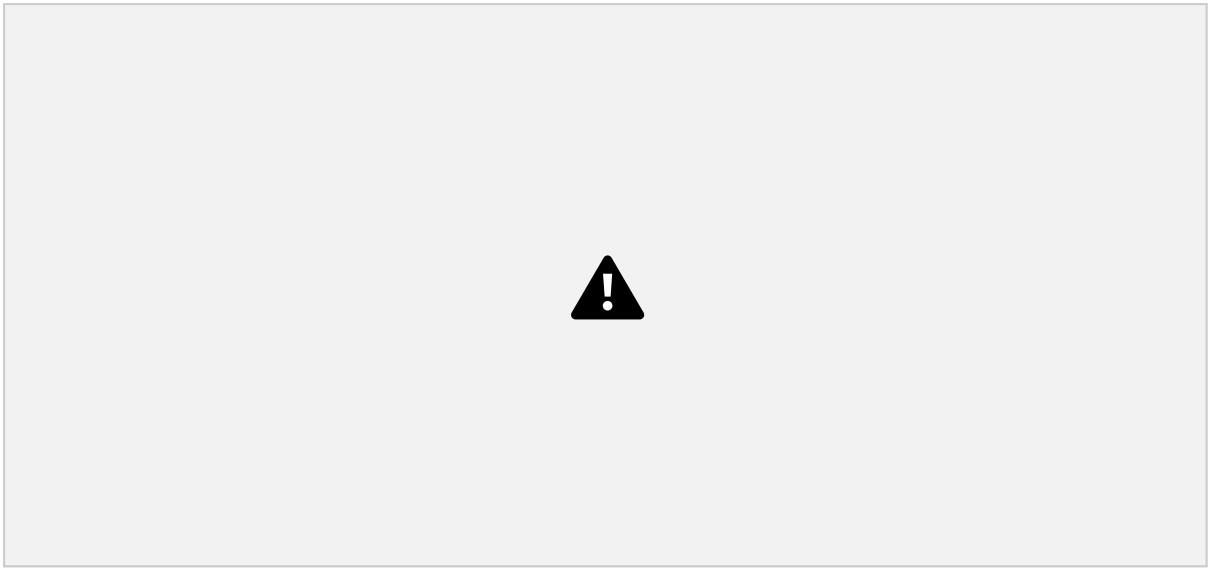
The COUNT() function counts the number of records.



SUM() FUNCTION

Purpose

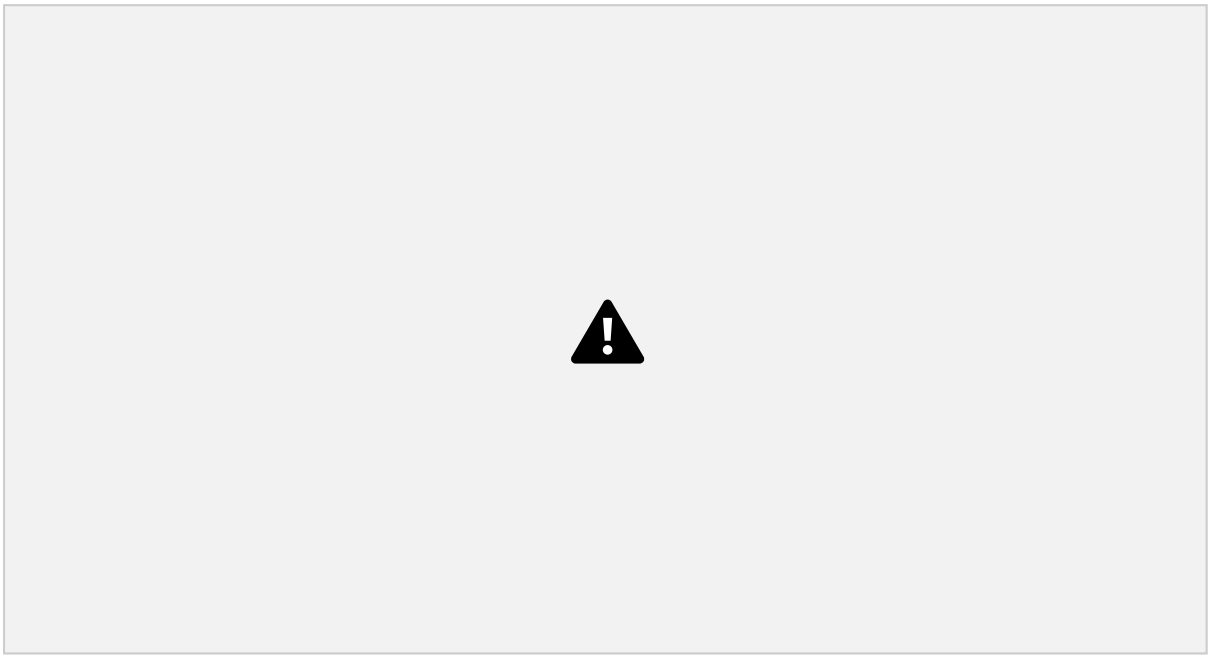
The SUM() function calculates the total of numerical values.



AVG() FUNCTION

Purpose

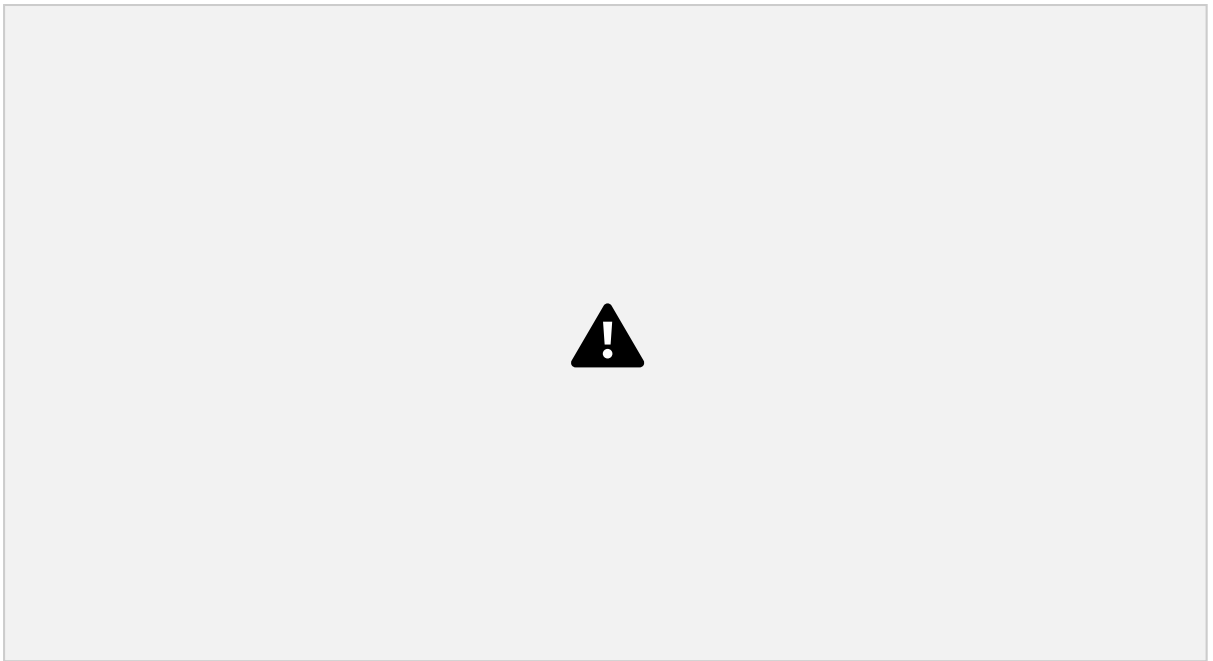
The AVG() function calculates the average of numerical values.



MAX() FUNCTION

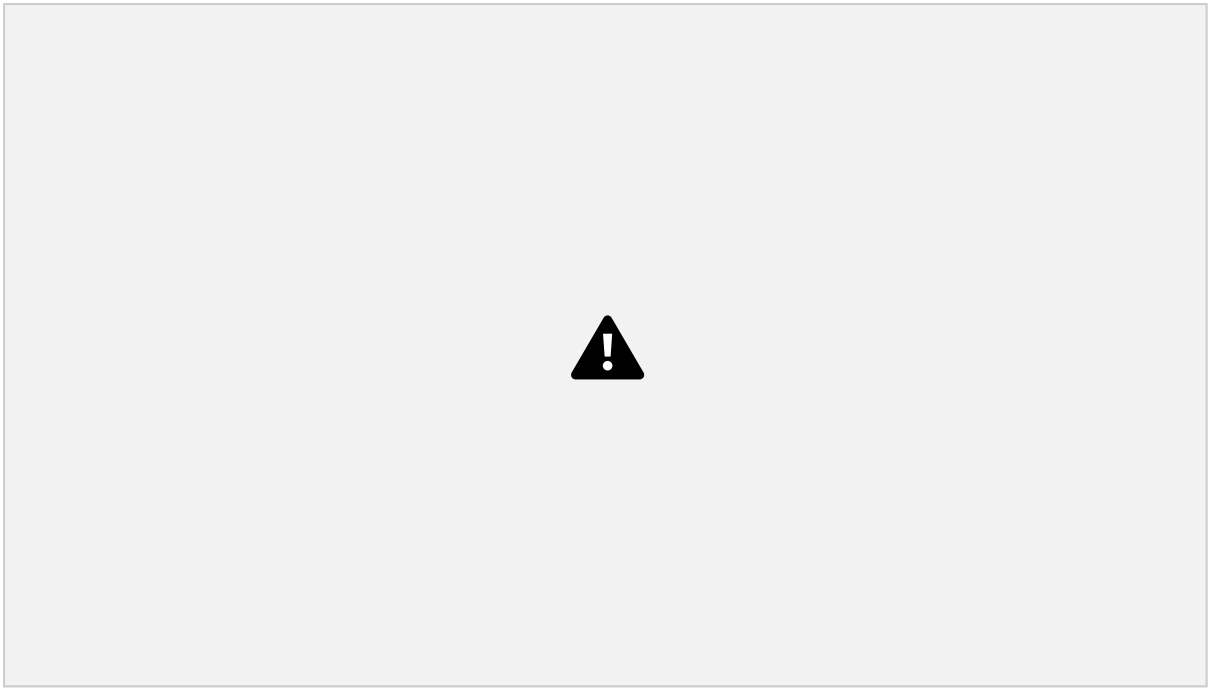
Purpose

The MAX() function finds the highest value from a column.



MIN() FUNCTION

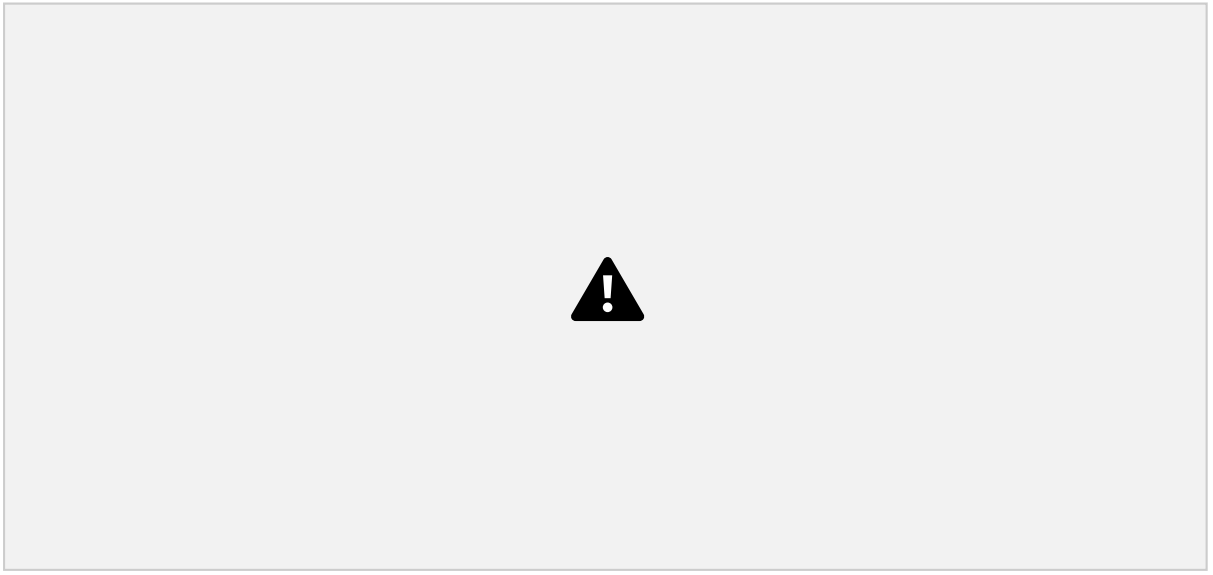
Purpose
The MIN() function finds the lowest value from a column.



AGGREGATE FUNCTIONS USING GROUP BY

Purpose

The GROUP BY clause is used to group records based on a column. Here, students are grouped according to their class, and the number of students and average marks are calculated for each class.

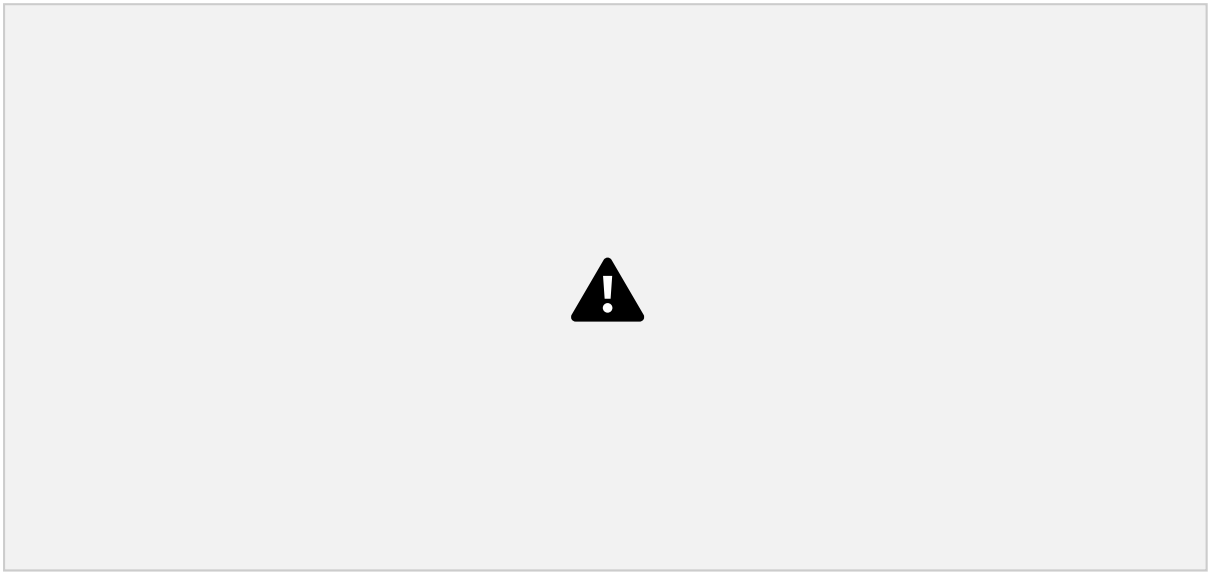


COMBINED SQL FUNCTIONS

Purpose

This query combines multiple SQL functions to generate a student report containing the student's name in uppercase, class, marks, difference from the target mark of 80, and birth

year.



CONCLUSION

The SQL functions were successfully applied to the STUDENT database. String functions were used to manipulate text data, Numeric functions were used to perform calculations, Date functions were used to extract and calculate date-related information, and Aggregate functions were used to perform calculations on multiple records. The GROUP BY clause was also used to obtain class-wise results, and a combined query was executed using multiple SQL functions.